

Evaluating the Impact of Post-Training Quantization on Inference Throughput and Accuracy in Gemma 4 e2B



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Research Objectives

- Measure generation speed (TPS) and Time to First Token (TTFT) across quantization levels (BF16 vs. 4-bit).
- Quantify the "Intelligence Loss" under PTQ using a zero-shot logical accuracy evaluation benchmark.
- Evaluate CPU and GPU loads during local edge inference to characterize hardware offload behaviors.

Research Implications

Implication 1

4-bit quantization is highly effective for accelerating inference throughput and saving VRAM on consumer GPUs.

Implication 2

Offloading parts of a model to CPU (due to memory overflow) triggers a massive performance cliff and high thermal loads.

Implication 3

Quantized models are not recommended for high-precision logic tasks due to structural weight-precision loss.

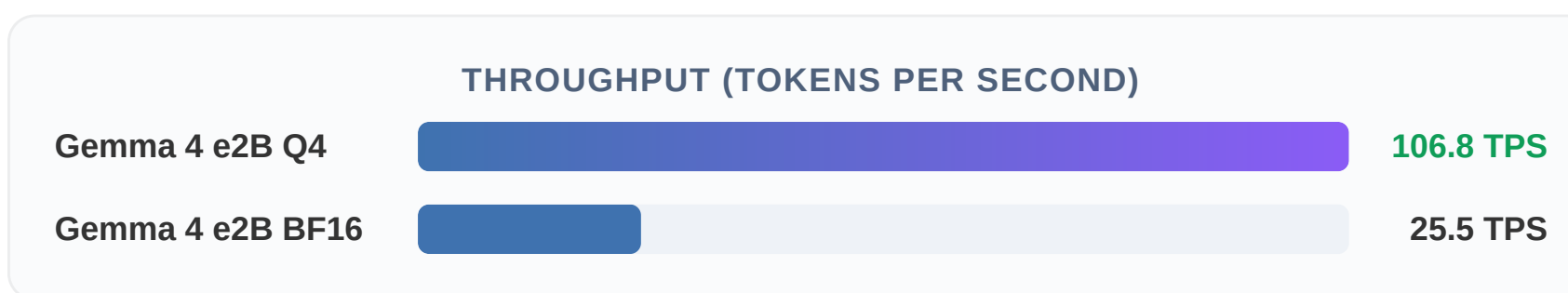
Implication 4

While Q4 works well for casual conversational chat, it exhibits critical errors in spatial reasoning and bitwise math.

KEY RESULTS

Performance Metrics: Throughput

Tokens Per Second (TPS) results show a massive performance gain when the model executes entirely on the GPU:



4-bit quantization yields a **4.2x speedup** by fitting the entire model (7.7 GB) inside the GPU's VRAM.

Hardware Telemetry & Resource Utilization

Actual system telemetry captured during active inference (derived from btop):

Telemetry Metric	Gemma Q4	Gemma BF16
CPU Core Utilization	14%	28%
CPU Package Temp	63°C	82°C
CPU Package Power Draw	22.9 W	34.9 W
GPU0 Core Utilization	85%	37%
GPU0 Core Temperature	56°C	68°C
GPU0 Core Power Draw	57.0 W	34.6 W
GPU0 VRAM Allocated	7.4 GB	4.9 GB (offloaded)
System RAM Allocation	4.42 GB	10.6 GB

Offloading parts of the 11 GB BF16 model to CPU memory creates bus bottlenecking, limiting GPU utility and raising CPU temps to **82°C**.

Summary

- 4-bit quantization (Q4) delivers 4.2x faster inference and fits inside GPU VRAM, ideal for edge systems.
- The uncompressed model (BF16) preserves 90% accuracy (vs Q4 80%), but suffers severe lag due to CPU offloading.

Background

- LLM transition from cloud APIs to local edge deployment requires compression to overcome GPU VRAM limits.
- Weight quantization (PTQ) significantly reduces VRAM footprint but risks degrading logical coherence.
- This study benchmarks uncompressed bfloat16 against 4-bit quantized GGUF variants to find the optimal trade-off.

Testing Dataset

Inference & Validation Dataset:

A logic evaluation suite of 10 zero-shot multiple-choice questions designed to isolate cognitive reasoning gaps under quantization. Answers are graded binarily (0 or 1) to measure absolute accuracy.

Methodology

Research Workflow:

Gemma 4 e2B Models → Load via Ollama (BF16 & Q4) → Execute 10 Zero-Shot Logic Puzzles → Log Telemetry (btop & ollama ps) → Evaluate Speed & Accuracy.

Variable	Type	Description
Quantization Bits	Independent	BF16 vs. Q4_K_M GGUF
Throughput (TPS)	Dependent	Tokens per second generated
Logic Accuracy	Dependent	Benchmark test score (0-10)
GPU Specs	Control	Linux OS, Dedicated 8GB VRAM

Zero-Shot Accuracy Evaluation

No	Logical Domain	Correct Answer	BF16	Q4
Q1	State Tracking (Coin)	A	A	A
Q2	Knight/Knave/Spy	B	B	B
Q3	Modular Arithmetic	B	B	B
Q4	Recursive Bitwise	C	C	C*
Q5	Cheryl's Birthday	B	A	A
Q6	Causal/Counterfactual	B	B	B
Q7	3D Die Rotation	C	C	A
Q8	Game Theory SPNE	C	C	C
Q9	Linguistic Negation	B	B	B
Q10	Bayesian Probability	A	A	A

*Note: Q4 model arrived at the correct answer in Q4 via flawed bitwise calculations (16 & 8 = 8), only succeeding due to option constraints.

Benchmark Accuracy Summary

Model Variant	Right (Correct)	Wrong (Incorrect)	Score
Gemma 4 e2B BF16	9	1	90%
Gemma 4 e2B Q4	8	2	80%

References

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